



HeatWatch 3 Industrial Temperature Telemetry System

Comprehensive Product Manual, Hardware Architecture, Operations & Troubleshooting Guide

Product Version: v3.0.4	Hardware Target: PPI AIME 8U / Pi 5	Target Host: goosepi (192.168.1.140)
Document Class: Technical & Operational Manual	OS Compatibility: Linux Bookworm / Raspberry Pi OS	Date: July 2026

1. System Overview & Purpose

HeatWatch 3 is an enterprise-grade, real-time industrial temperature telemetry and alarm management solution engineered for 24/7 continuous process monitoring. Interfacing directly with the **PPI AIME 8U 8-channel RTD Modbus hardware module**, HeatWatch 3 provides ultra-reliable, sub-second temperature acquisition across critical thermal loops including industrial boilers, heat exchangers, chilled water loops, storage tanks, and ambient plant environments.

Built on a resilient microservice architecture, the system combines hardware Modbus polling (`poller.py`), high-throughput InfluxDB time-series logging, Node.js WebSocket broadcasting, and a zero-dependency, touch-optimized HTML5 frontend tailored for 7" and 10" Raspberry Pi kiosk displays.

2. Core Product Features

Feature Feature Category	Technical Capabilities & Specifications
8-Channel Real-time Grid	Live telemetry updated every 2 seconds via WebSockets. Includes channel badge, sensor label, progress bar, threshold markers, and dynamic 'Last updated X sec ago' ticker.
5-Tier Visual Alarm System	• HIHI ALARM / LOLO ALARM: Solid vivid red card fill (#dc2626) with pulsing glow & SCADA chime. • WARNING HI / WARNING LO: Solid bright golden yellow card fill (#eab308) with amber glow. • NORMAL: Clean slate container with green status indicator. • OFFLINE: Grayed-out state for disconnected probes.
SCADA Audio Siren	Web Audio SCADA dual-tone emergency chime (960Hz → 720Hz & 1200Hz → 840Hz chirps) with header mute toggle. Unlocks automatically on user gesture.
Temperature Unit Conversion	Instant system-wide unit switching between Celsius (°C) , Fahrenheit (°F) , and Kelvin (K) across live cards, trend graphs, 24h stats, and historical export logs.
Time-Series Trends Chart	Multi-channel Chart.js trend curves supporting 1h, 6h, and 24h historical windows, high-luminance dark mode colors, axis scale autoscale, and tooltip inspection.
24-Hour Channel Statistics	Min, Avg, and Max daily summary cards calculated dynamically per channel over rolling 24-hour log windows.
Fullscreen Kiosk & Virtual Keyboard	One-click native fullscreen kiosk toggle and built-in QWERTY/Numpad virtual keyboard for standalone Pi touchscreen administration without external peripherals.
Branded Data Exports	One-click generation of Goose-branded PDF telemetry reports, formatted Excel workbooks (.xlsx), and raw CSV logs.
System Health Diagnostics	Monitors Pi CPU Load %, CPU Temp (vcgencmd), RAM usage, NVMe/SD Disk %, InfluxDB size, system uptime, and PPI AIME 8U hardware connection badge.

3. Hardware & Software Architecture

HeatWatch 3 utilizes a decoupled pipeline architecture designed for high availability and data integrity:

- **Hardware Layer (PPI AIME 8U):** 8 3-wire RTD PT100 sensors connected to the PPI Modbus RTU module at IP 192.168.1.2 (Port 502 / RS485).
- **Ingestion Daemon (poller.py):** Python background daemon executing Modbus read queries every 2 seconds. Stores valid readings in **InfluxDB v2** bucket heatwatch_telemetry.
- **Web & Application Server (server.js):** Node.js Express server listening on Port 3001. Serves frontend assets, manages REST API endpoints, queries InfluxDB flux data, and broadcasts WebSocket JSON frames.
- **Frontend Dashboard (public/index.html):** Zero-dependency Single Page Application (SPA) utilizing HTML5, Vanilla CSS, Chart.js, and Web Audio API.

4. Step-by-Step Operating Guide

4.1 Initial Setup Wizard

Upon first system launch, HeatWatch 3 displays an interactive 4-step setup wizard:

1. **Welcome Screen:** Overview of PPI AIME 8U integration.
2. **Administrator Account:** Set Admin Username, Password, Role, and 4-digit Security Recovery PIN.
3. **RTD Channel Configuration:** Define channel names and set threshold limits (LoLo, Lo, Hi, HiHi) for all 8 channels on a zero-scroll table.
4. **Review & Launch:** Confirm settings to initialize the telemetry dashboard.

4.2 Live Telemetry Grid & Alarms

The Live view presents 8 channel cards updating in real time. If a sensor temperature breaches set limits:

- **Warning (Hi/Lo):** The entire card container pulses in bright golden yellow (#eab308).
- **Alarm (HiHi/LoLo):** The entire card pulses in vivid solid red (#dc2626) and triggers the SCADA emergency audio chime.
- **Muting Sound:** Click the speaker icon in the top navbar to silence the audio alarm.

4.3 Viewing Trends & Exporting Logs

Navigate to the **Trends** tab to analyze historical curves and 24-hour channel summary cards. Use the **History** tab to query past logs by time window (1h/6h/24h/7d) or channel filter, and click **PDF** or **Excel** to export reports.

4.4 Settings & Protected Administration

Open the Settings panel (gear icon) to toggle **Temperature Units (°C/°F/K)** or **Hide Offline Channels**. Protected settings (editing channel labels or thresholds) require entering your Admin password.

5. Network & System Administration

To ensure permanent control room access, HeatWatch 3 includes dedicated network configuration utilities:

5.1 Static IP Setup (`192.168.1.140`)

Lock the Pi IP to `192.168.1.140` using the provided automated script:

```
cd ~/Heatwatch-Version-3 && sudo bash scripts/setup-static-ip.sh 192.168.1.140/24 192.168.1.1
```

5.2 Local Hostname (mDNS)

With Avahi mDNS enabled, users can connect to the dashboard via `http://heatwatch.local:3001` or `http://goosepi.local:3001` from any device on the local network.

5.3 Systemd Service Management

The system runs as an auto-starting background service `heatwatch-dashboard.service`:

```
# Check Status: sudo systemctl status heatwatch-dashboard # Restart Service: sudo systemctl restart heatwatch-dashboard
```

6. Comprehensive Troubleshooting Matrix

Symptom / Issue	Probable Cause	Step-by-Step Resolution
Sensor card displays 0.0 or OFFLINE	1. RTD probe disconnected. 2. PPI hardware offline. 3. Poller daemon stopped.	1. Inspect physical 3-wire RTD probe wiring at PPI terminal block. 2. Check PPI badge in Diagnostics tab. 3. Restart service: <code>sudo systemctl restart heatwatch-dashboard</code>.
PPI Badge displays 'PPI Offline'	PPI AIME 8U module at IP 192.168.1.2 unreachable or powered off.	1. Verify 24V DC power supply to PPI module. 2. Ping PPI module: <code>ping 192.168.1.2</code>. 3. Check Ethernet/RS485 cable between Pi and PPI.
Dashboard URL unreachable via Browser	1. Pi IP changed. 2. Node.js server crashed.	1. Verify Pi IP: <code>hostname -I</code>. 2. Re-apply static IP: <code>sudo bash scripts/setup-static-ip.sh 192.168.1.140/24 192.168.1.1</code>. 3. Check logs: <code>journalctl -u heatwatch-dashboard -f</code>.
Alarm SCADA Audio not playing	1. Browser audio muted. 2. Autoplay policy blocking audio.	1. Check speaker mute button in top navbar. 2. Click anywhere on the browser window to unlock Web Audio API permission.
Touchscreen Keyboard not appearing	Input focus event lost or touch listener suppressed.	Tap directly into an input field (username/password/threshold) to activate the onscreen QWERTY virtual keyboard.
InfluxDB Connection / Log Query Error	InfluxDB service inactive or database token expired.	1. Check InfluxDB service: <code>sudo systemctl status influxdb</code>. 2. Restart InfluxDB: <code>sudo systemctl restart influxdb</code>.